COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE009/Data Structures and Algorithms	**SECTION:**	CPE21S3
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HOA 5.1 Queues

1. PROCEDURE OUTPUT

ILO A:

Code:

```cpp
#include <iostream>
#include <queue>

void display(std::queue<char> copyQ);

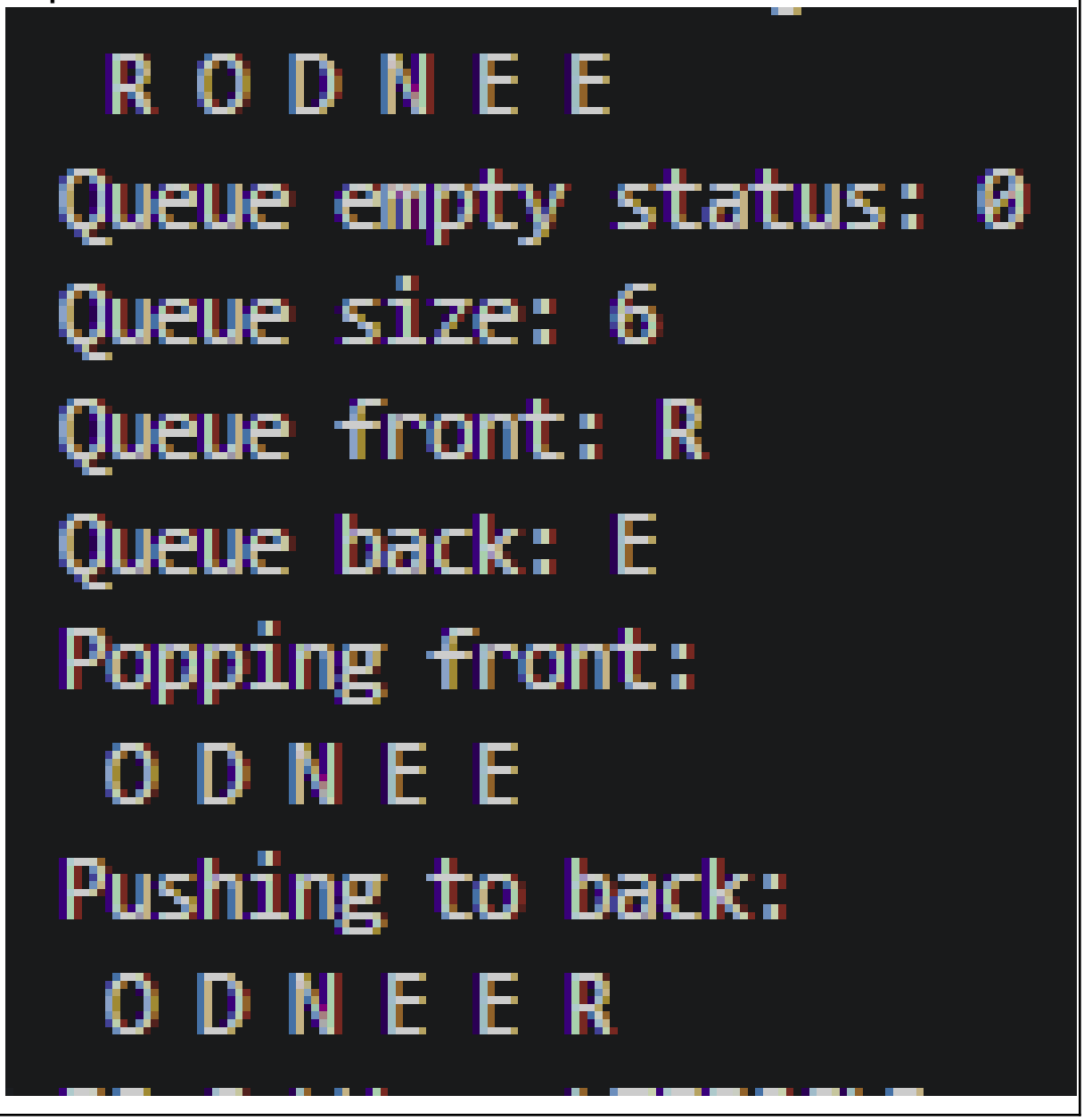
int main(){
    std::queue<char> myQ;
    myQ.push('R');
    myQ.push('O');
    myQ.push('D');
    myQ.push('N');
    myQ.push('E');
    myQ.push('E');
    display(myQ);

    std::cout<<"Queue empty status: "<<myQ.empty()<<std::endl;
    std::cout<<"Queue size: "<<myQ.size()<<std::endl;
    std::cout<<"Queue front: "<<myQ.front()<<std::endl;
    std::cout<<"Queue back: "<<myQ.back()<<std::endl;
    std::cout<<"Popping front: "<<std::endl;
    myQ.pop();
    display(myQ);
    std::cout<<"Pushing to back: "<<std::endl;
    myQ.push('R');
    display(myQ);

    return 0;
}

void display(std::queue<char> copyQ){
    std::queue<char> c=copyQ;
    while(!c.empty()){
        std::cout<<" "<<c.front();
        c.pop();
    }
    std::cout<<std::endl;
}
```

Output:


R O D N E E
Queue empty status: 0
Queue size: 6
Queue front: R
Queue back: E
Popping front:
O D N E E
Pushing to back:
O D N E E R

ILO B:

Code:

main.py:

```cpp
#include <iostream>
#include "queueLL.h"

int main(){
    qNode<char>* front=nullptr;
    qNode<char>* back=nullptr;

    std::cout<<"Enqueuing: "<<std::endl;
    enqueue('R',&front,&back);
    std::cout<<front->data<<" "<<back->data<<std::endl;
    enqueue('O',&front,&back);
    std::cout<<front->data<<" "<<back->data<<std::endl;
    enqueue('D',&front,&back);
    std::cout<<front->data<<" "<<back->data<<std::endl;
    enqueue('N',&front,&back);
    std::cout<<front->data<<" "<<back->data<<std::endl;
    enqueue('E',&front,&back);
    std::cout<<front->data<<" "<<back->data<<std::endl;
    enqueue('E',&front,&back);
    std::cout<<front->data<<" "<<back->data<<std::endl;

    std::cout<<"Front: "<<std::endl;
    checkFront(front);

    std::cout<<"Back: "<<std::endl;
    checkBack(back);

    std::cout<<"Display all: "<<std::endl;
    displayAll(&front,&back);
    std::cout<<std::endl;

    std::cout<<"Dequeuing: "<<std::endl;
    dequeue(&front,&back);
    std::cout<<front->data<<" "<<back->data<<std::endl;
    dequeue(&front,&back);
    std::cout<<front->data<<" "<<back->data<<std::endl;
    dequeue(&front,&back);
    std::cout<<front->data<<" "<<back->data<<std::endl;

    std::cout<<"Front: "<<std::endl;
    checkFront(front);

    std::cout<<"Back: "<<std::endl;
    checkBack(back);


    return 0;
}
```

queueLL.h:

```
#ifndef QUEUELL_H
#define QUEUELL_H

template <typename T>
class qNode{
    public:
        T data;
        qNode<T>* next;

};

template <typename T>
qNode<T>* new_node(T newData){
    qNode<T>* newNode=new qNode<T>;
    newNode->data=newData;
    newNode->next=nullptr;
    return newNode;
};

template <typename T>
void enqueue(T newData,qNode<T>** frontPtr,qNode<T>** backPtr){
    qNode<T>* newNode=new_node(newData);

    if((*frontPtr)==nullptr&&(*backPtr)==nullptr){
        (*frontPtr)=newNode;
        (*backPtr)=newNode;
        return;
    }

    (*backPtr)->next=newNode;
    (*backPtr)=newNode;
}

template <typename T>
void dequeue(qNode<T>** frontPtr,qNode<T>** backPtr){
    qNode<T>* deleteNode=nullptr;
    deleteNode=(*frontPtr);

     if((*frontPtr)==nullptr&&(*backPtr)==nullptr){
        std::cout<<"The queue is empty\n";
        return;}

    if((*frontPtr)->next==nullptr&&(*backPtr)->next==nullptr){
        (*frontPtr)=nullptr;
        (*backPtr)=nullptr;
        delete deleteNode;
    }
```

```
49      (*frontPtr)=(*frontPtr)->next;
50      deleteNode->next=nullptr;
51      delete deleteNode;
52  }
53
54  template<typename T>
55  void checkFront(qNode<T>* frontPtr){
56      if((frontPtr)==nullptr) return;
57
58      std::cout<<frontPtr->data<<std::endl;
59  }
60
61  template<typename T>
62  void checkBack(qNode<T>* backPtr){
63      if((backPtr)==nullptr){
64          std::cout<<"Front is empty"<<std::endl;
65          return;
66      }
67      std::cout<<backPtr->data<<std::endl;
68  }
69
70  template<typename T>
71  void displayAll(qNode<T>** frontPtr,qNode<T>** backPtr){
72      if((*frontPtr)==nullptr&&(*backPtr)==nullptr){
73          std::cout<<"Queue is empty"<<std::endl;
74          return;
75      }
76      qNode<T>* displayNode=nullptr;
77      displayNode=(*frontPtr);
78
79      while(displayNode->next!=nullptr){
80          std::cout<<displayNode->data;
81          displayNode=displayNode->next;
82      }
83      std::cout<<displayNode->data<<std::endl;
84  }
85
86  #endif
```

Output:

```
Enqueuing:
R R
R O
R D
R N
R E
R E
Front:
R
Back:
E
Display all:
RODNEE

Dequeuing:
O E
D E
N E
Front:
N
Back:
E
```

ILO A Observations:

Similar to the stack, it hosts a number of functions like pop and push, and like the stack, it tracks the last in, but also the first in. Like the characters that spell RODNEE, R is popped out, and R is pushed back in, but on the other end.

ILO B Observations:

In enqueuing, it tracks the first in and last in. Adding R makes it the first and last in line. But as more characters are queued, the back or end character changes. Displaying all in the queue requires a loop that prints out each character until it reports null, thus spelling my name like a string. In dequeuing, it removes the first character in line, making the second become first in line and so on until the dequeuing stops. The front displays the first character in line, while the back prints the last one.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Grammarly**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

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